Figure 18.5 Various input element types. The exact way in which each type is displayed varies by browser.

To handle this diversity, the Input Tag Helper is one of the most powerful Tag Helpers. It uses information based on both the type of the property (`bool`, `string`, `int`, and so on) and any `DataAnnotations` attributes applied to it (`[EmailAddress]` and `[Phone]`, among others) to determine the type of the `input` element to generate. The `DataAnnotations` are also used to add `data-val-*` client-side validation attributes to the generated HTML.

Consider the `Email` property from listing 18.2 that was decorated with the `[EmailAddress]` attribute. Adding an `<input>` is as simple as using the `asp-for` attribute:

```
<input asp-for="Input.Email" />
```

The property is a `string`, so ordinarily the Input Tag Helper would generate an `<input>` with `type="text"`. But the addition of the `[EmailAddress]` attribute provides additional metadata about the property. Consequently, the Tag Helper generates an HTML5 `<input>` with `type="email"`:

```
<input type="email" id="Input_Email" name="Input.Email"
    value="test@example.com" data-val="true"
    data-val-email="The Email Address field is not a valid e-mail address."
    Data-val-required="The Email Address field is required."
    />
```

You can take a whole host of things away from this example. First, the `id` and `name` attributes of the HTML element have been generated from the name of the property. The value of the `id` attribute matches the value generated by the Label Tag Helper in its `for` attribute, `Input_Email`. The value of the `name`